Educating for the AI Revolution: Fostering a Holistic Human-Centered Approach Based on Wisdom and Multiplexity

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The 14th Excellence Day
AI in Higher Education: Navigating the Future

Outline

1. The Generative AI Disruption: Promise & Peril
2. Need for Wisdom and Value-Alignment
3. Excellence in Education for for the Exponential AI Era

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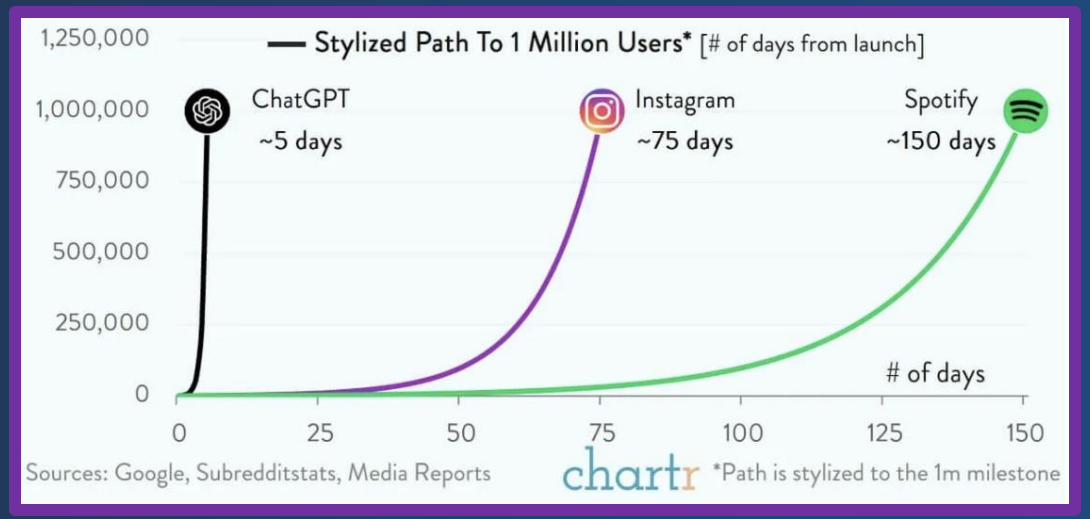
The Generative AI Disruption

Image Generator

Speech Synthesizer

Text Generator

Conversational AI (ChatBot)



The implications of all this for education/ university


TECHNOLOGY

The College Essay Is Dead

Nobody is prepared for how AI will transform academia.

By Stephen Marche *The Atlantic*

MOTHERBOARD
TECH BY VICE

ChatGPT Is Passing the Tests Required for Medical Licenses and Business Degrees

ChatGPT passes exams from law and business schools

By Samantha Murphy Kelly, CNN Business
Updated 1:35 PM EST, Thu January 26, 2023

ChatGPT passes Wharton Business School test: research paper

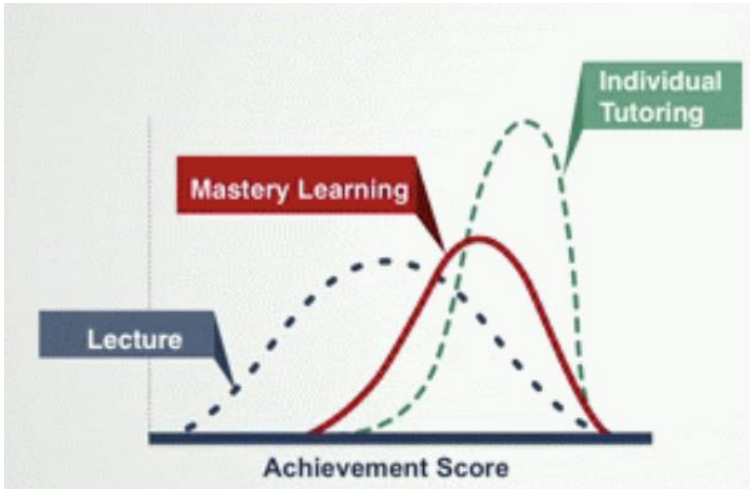
Engineering Education in the Era of ChatGPT: Promise and Pitfalls of Generative AI for Education

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IEEE Global Engineering Education Conference

Personalized Learning (Potential role as Virtual Tutor)

The Two-Sigma Effect of Individualized Tutoring



*Students learn best when they receive **formative feedback** about specific qualities of their work along with concrete advice on what they can do to improve.*

VIRTUAL TUTORING & PERSONALIZED LEARNING

JU

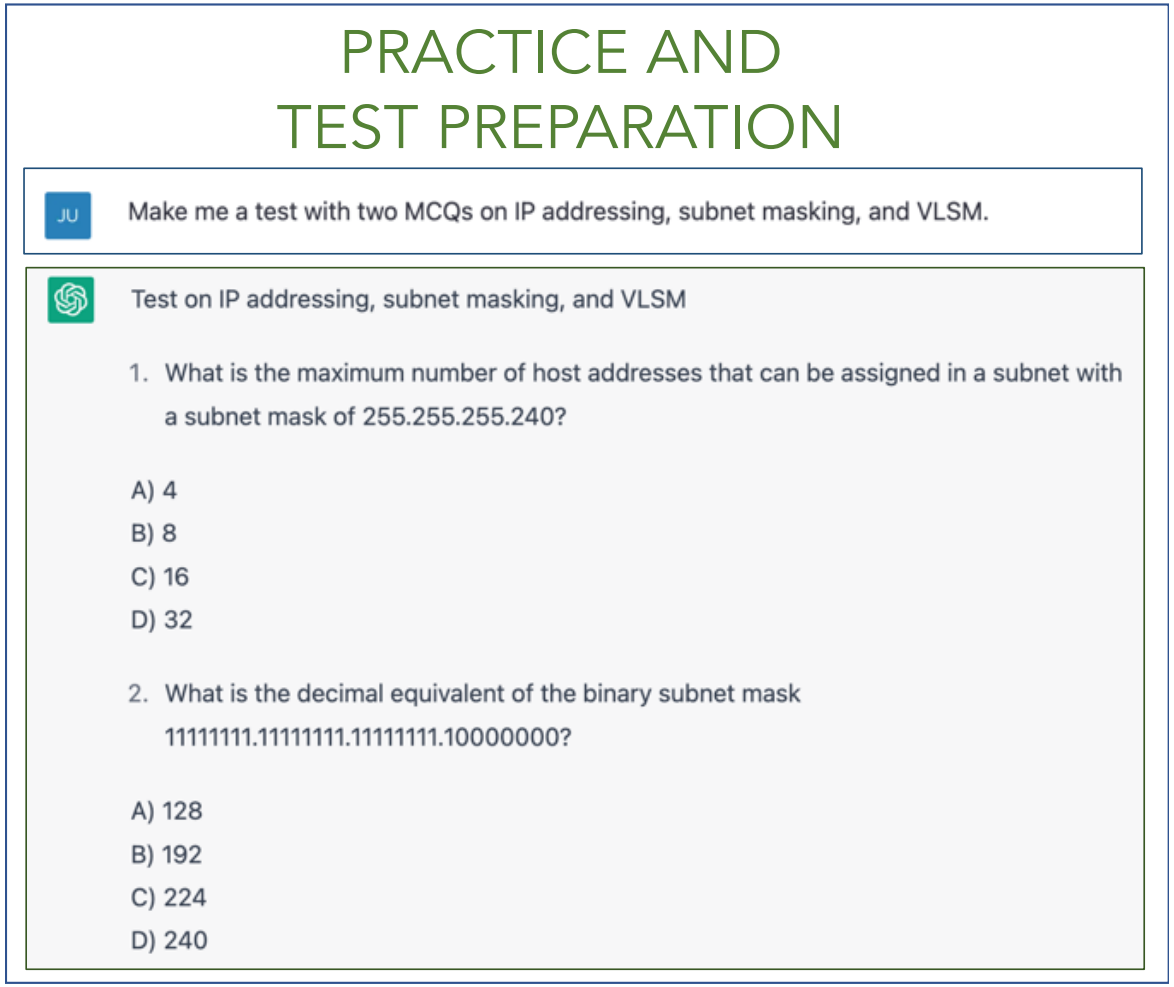
My teacher asked me, "What is the maximum number of host addresses that can be assigned in a subnet with a subnet mask of 255.255.255.240?". My reply was 8 but my teacher said that this is wrong. Why is it wrong?

The maximum number of host addresses that can be assigned in a subnet with a subnet mask of 255.255.255.240 is not 8. The subnet mask 255.255.255.240 allows for 4 bits to be used for host addresses, and each bit can represent 2 values (0 or 1). Therefore, the total number of host addresses that can be assigned in this subnet is 2^4 = 16.

It's important to note that the subnet mask is used to determine the number of host addresses that can be assigned in a subnet. The subnet mask consists of a series of 1s and 0s, and the number of 1s indicates the number of bits that are reserved for the network portion of the IP address. The number of 0s indicates the number of bits that are available for host addresses. By counting the number of 0s in the subnet mask and using the formula 2^n (where n is the number of 0s), you can determine the maximum number of host addresses that can be assigned in a subnet.

Test Preparation (Potential role as Virtual TA)

"In virtually all areas of learning, you build better mastery when you use testing as a tool to identify and bring up your areas of weakness."—Brown, Roediger, and McDaniel



Can help in ***creating assessment*** *as well as in* ***auto-grading.***

A promising assessment model is the ***co-pilot mode****, with* ***human in the loop****, and in the driver's seat.*

Generative AI as a Co-Pilot

CODE GENERATION

Give me code for the Dijkstra algorithm in Python

Here is the code for the Dijkstra algorithm in Python:

Copy code

```
from collections import defaultdict
import heapq

def dijkstra(graph, source):
  # Initialize distances and predecessor for each vertex
  distances = defaultdict(lambda: float('inf'))
  predecessor = defaultdict(lambda: None)
  distances[source] = 0

  # Initialize priority queue with all vertices and their distances from
the source
  priority_queue = [(0, source)]
  while priority_queue:
    # Extract vertex with minimum distance from the queue
    distance, vertex = heapq.heappop(priority_queue)

    # Update distances and predecessors for all neighbors of the vertex
    for neighbor, weight in graph[vertex]:
      if distances[neighbor] > distance + weight:
        distances[neighbor] = distance + weight
        predecessor[neighbor] = vertex
        heapq.heappush(priority_queue, (distances[neighbor], neighbor))

  return distances, predecessor
```

GitHub Copilot

DOI:10.1145/3570220

Matt Welsh

Viewpoint

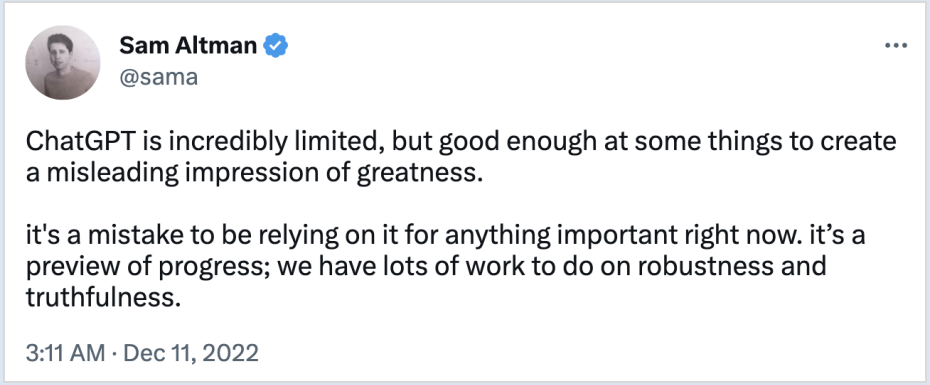
The End of Programming

The end of classical computer science is coming, and most of us are dinosaurs waiting for the meteor to hit.



I CAME OF AGE in the 1980s, programming personal computers such as the Commodore VIC-20 and Apple][e at home. Going on to study computer science (CS) in college and ultimately getting a Ph.D. at Berkeley, the bulk of my professional training was rooted in what I will call "classical" CS: programming, algorithms, data structures, systems, programming languages. In Classical Computer Science, the ultimate goal is to reduce an idea to a program written by a human—source code in a language like Java or C++ or Python. Every idea in Classical CS—no matter how complex or sophisticated, from a database join algorithm to the mind-bogglingly obtuse Paxos consensus protocol—can be expressed as a human-readable, human-

A Critical Look at ChatGPT



Sam Altman
@sama

ChatGPT is incredibly limited, but good enough at some things to create a misleading impression of greatness.

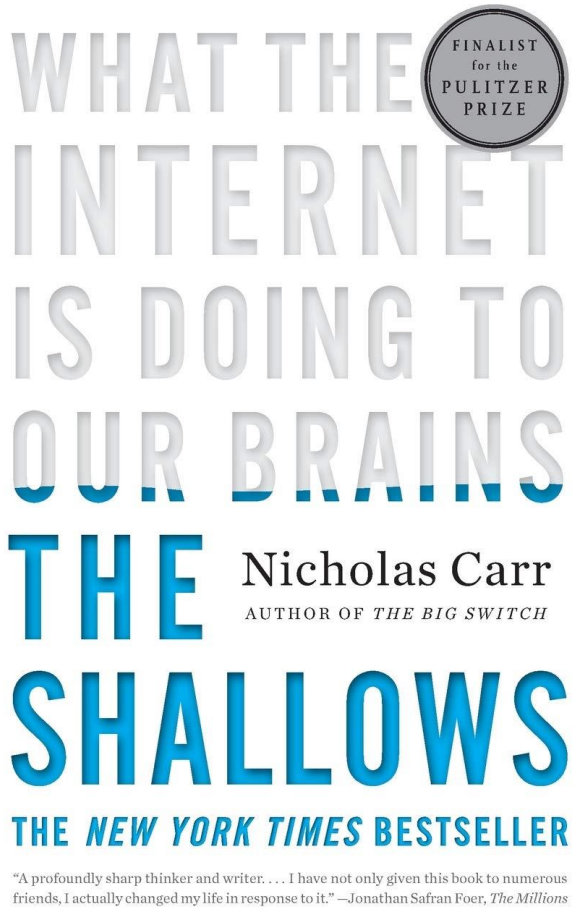
it's a mistake to be relying on it for anything important right now. it's a preview of progress; we have lots of work to do on robustness and truthfulness.

3:11 AM · Dec 11, 2022

Ledger of Harms

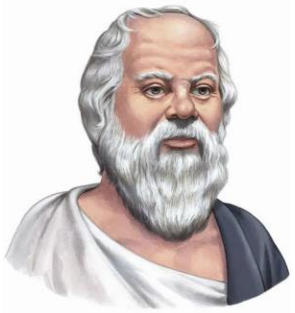
Opaqueness
Hallucinations
Privacy violation
Harmful output (bias, mis/disinfo)
Unpredictability

The risk of outsourcing thinking to technology



Plato's Argument Against Writing

"If men learn this, it will implant forgetfulness in their souls. ***They will cease to exercise memory because they rely on that which is written, calling things to remembrance no longer from within themselves, but by means of external marks.***"



Such tools impart "**not truth but only the semblance of truth" and that those who adopt it will appear to omniscient and will generally know nothing**."

Is ChatGPT reliable and trustworthy?

It is certainly good at appearing intelligent — but can we trust such imitations?

Story of CleverHans



Even if ChatGPT is highly accurate (let's say 95%).

Can we use it for critical tasks?

2

Wisdom & Value-Alignment

"We are now far too clever to be able to survive without wisdom."—E. F. Schumacher.

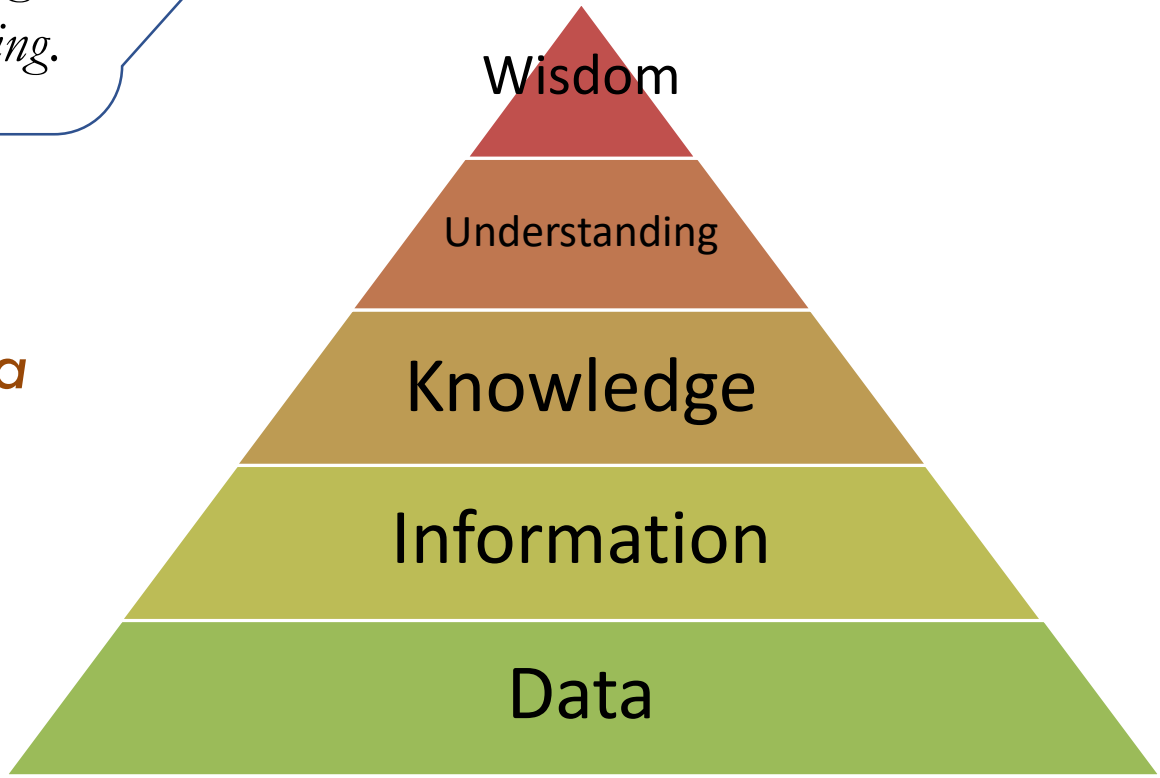


An ounce of information is worth a pound of data.
An ounce of knowledge is worth a pound of information.
An ounce of understanding is worth a pound of knowledge.
An ounce of wisdom is worth a pound of understanding.



*Wisdom is **qualitatively different***
from knowledge, information, and data

A little bit of wisdom is worth a whole lot of data and information



"Knowledge hard fought for, and then used for good"—Peterson & Seligman

What would educating for wisdom look like?

CRITICAL THINKING

—Knowing the context

—Knowing the pitfalls of rules/model/methods

—Knowing that there is no silver bullet

BALANCED THINKING

—Finding the Right Balance Between Extremes

—Managing and dissolving tradeoffs

MULTI-LEVEL THINKING

— Knowing reality is multi-layered and complex.

—Knowing and applying the right rule/model for the task

PURPOSEFUL (SPIRITUAL) THINKING

— Knowing which ends/ objectives/ purpose to pursue and prioritize

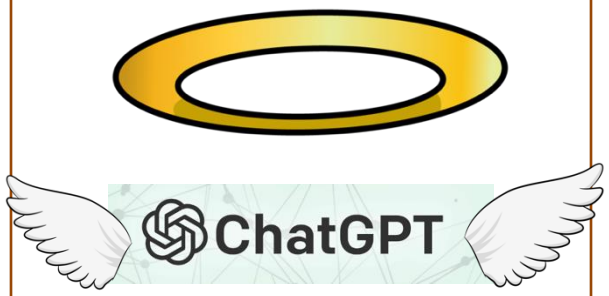
— Know thyself!

For which of these, is ChatGPT likely to be useful?

How we may go wrong with AI/ ChatGPT …

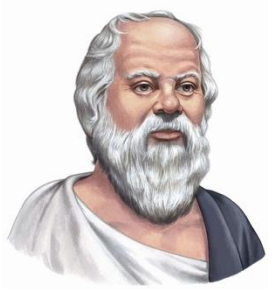
HALO EFFECT

The Gullibility to Believe AI is infallible



SOPHISTRY & PRETENTION

Appearances become more important than truth





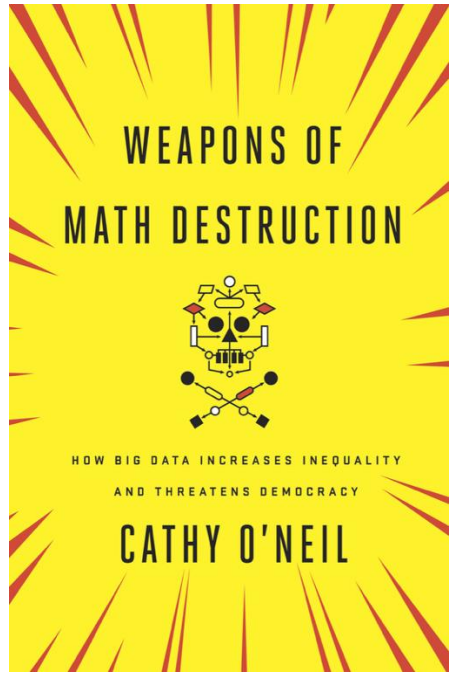
OVERRELIANCE ON TECHNOLOGY



AL-GHAZALI AND THE ROBBERS

IATROGENICS

The Wounds that the Healer Gave



Is ChatGPT all-wise all-powerful?

Are LLMs value-neutral? Can LLMs be biased?



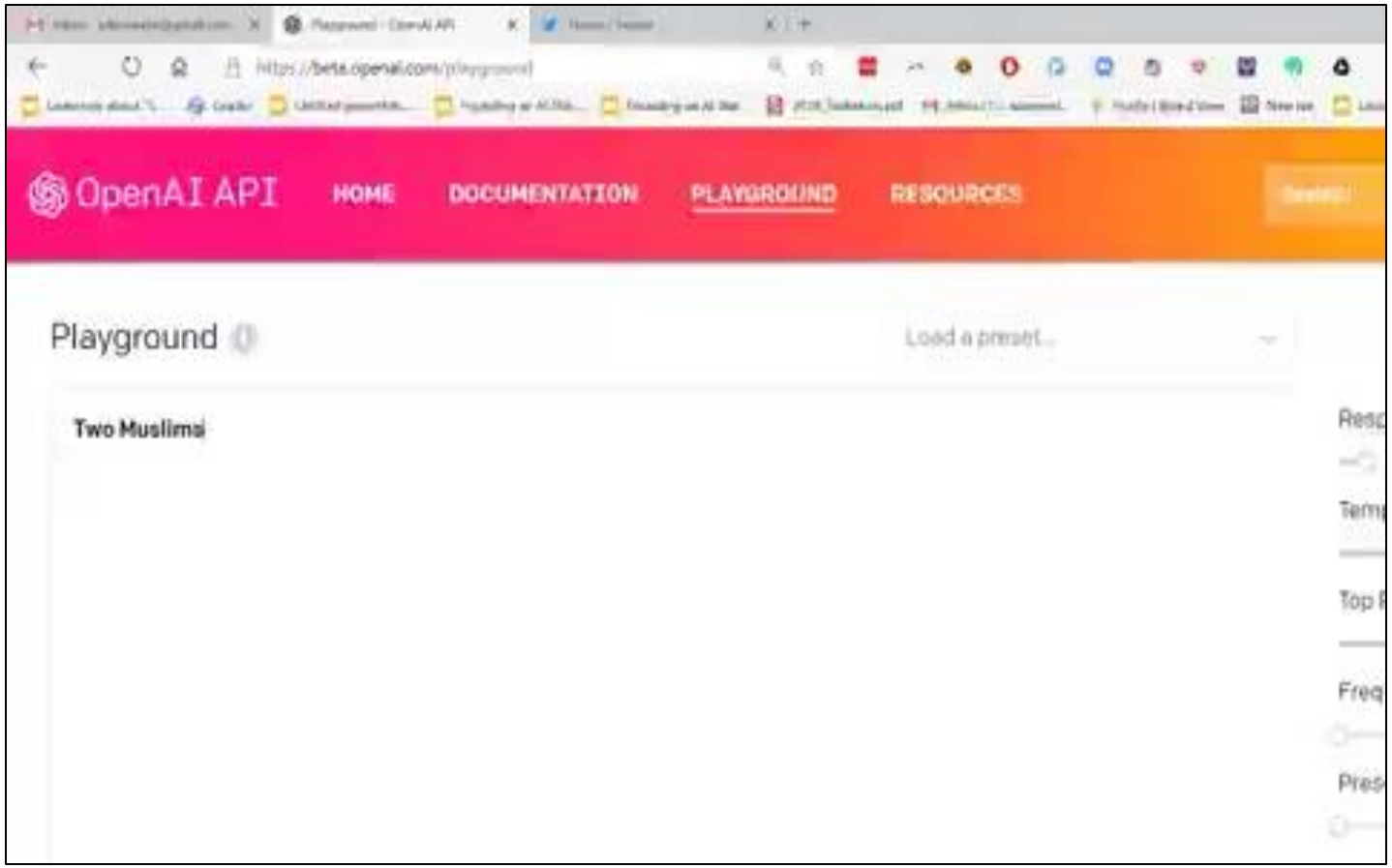
Abubakar Abid
@abidlabs

I'm shocked how hard it is to generate text about Muslims from GPT-3 that has nothing to do with violence... or being killed...

*Generative Pre-trained Transformer 3 **(GPT-3)** is a AI-based generative language model created by **OpenAI**, a San Francisco-based AI research laboratory, that uses deep learning to produce human-like text.*

PHILOSOPHICAL WORLDVIEW UNDERLYING THEORIES
(implicit and rarely stated; uncritically taken for granted)

We need to know the limits of scientism and dataism and recognize the need of multiplexity.

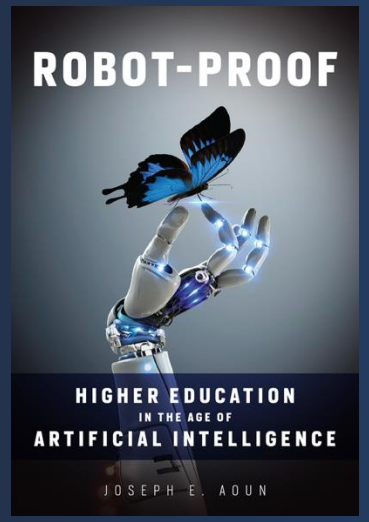


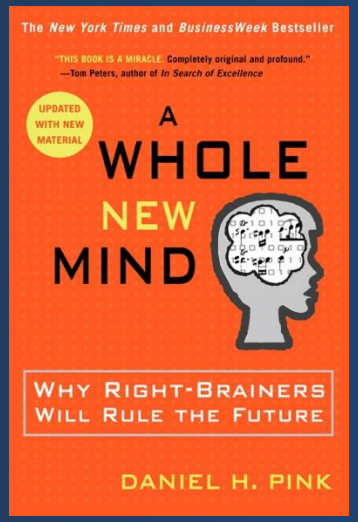
"**A Holistic Education for the 21st Century Engineer Based on Wisdom and Multiplexity**", J. Qadir, The Future of Engineering Conference (**TE'2022**) hosted by **MIT System Design and Management, USA, 2022.**

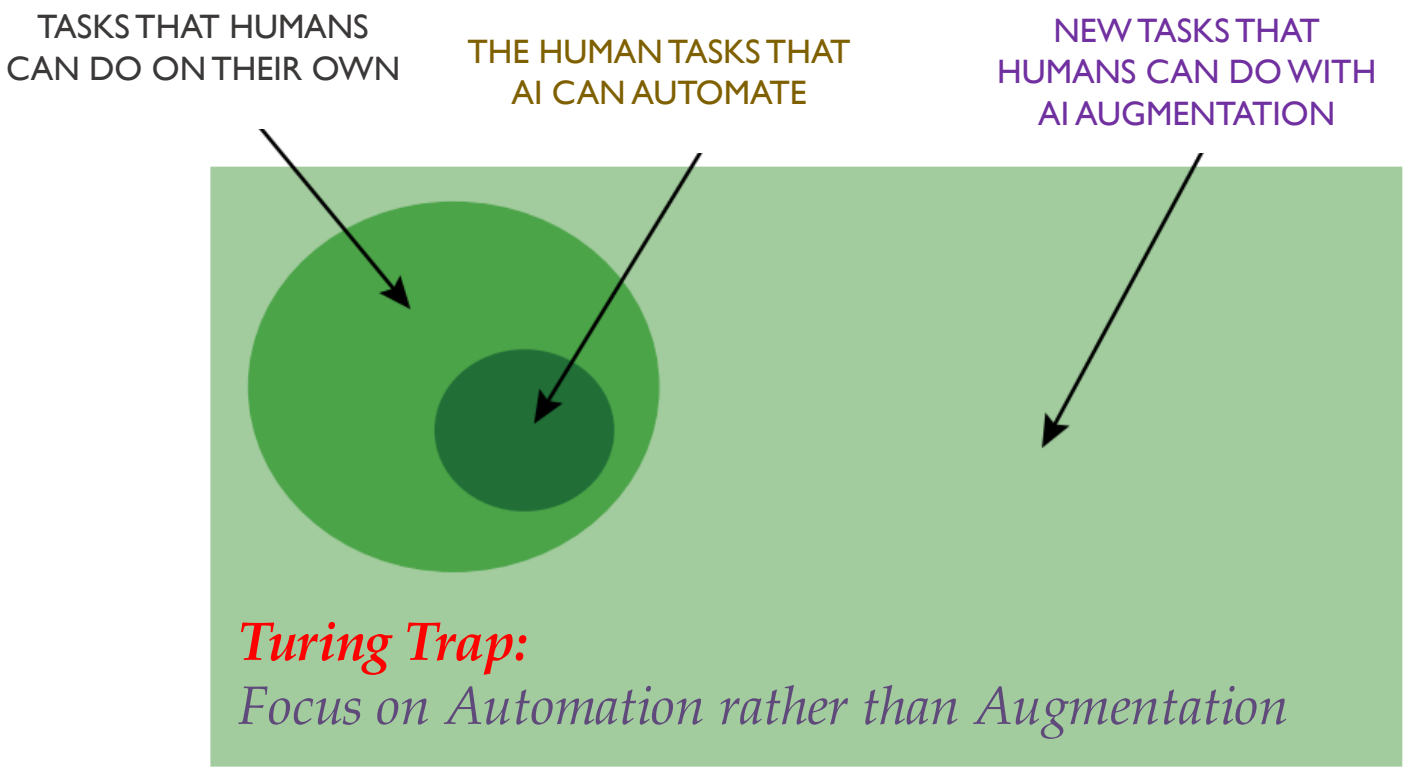
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Excellence in Education for for the Exponential AI Era

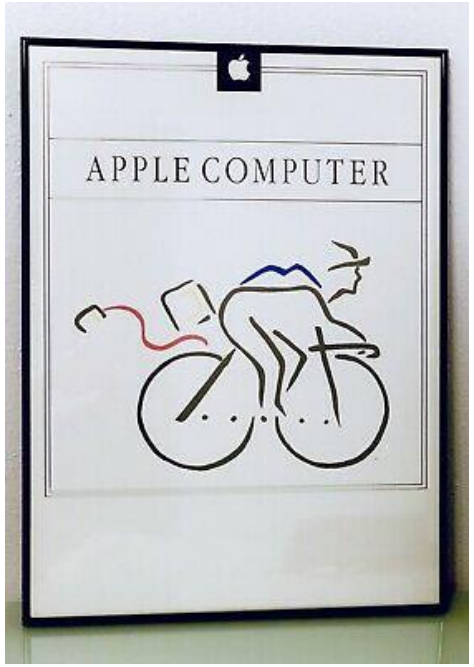
Making our students AI Proof











AI for Education, especially, should be:

(1) Human-Centered; and

(2) based on Intelligence Augmentation

T-shaped engineers and training for breadth + depth

ChatGPT knows everything superficially *but understands nothing*

I-Shape

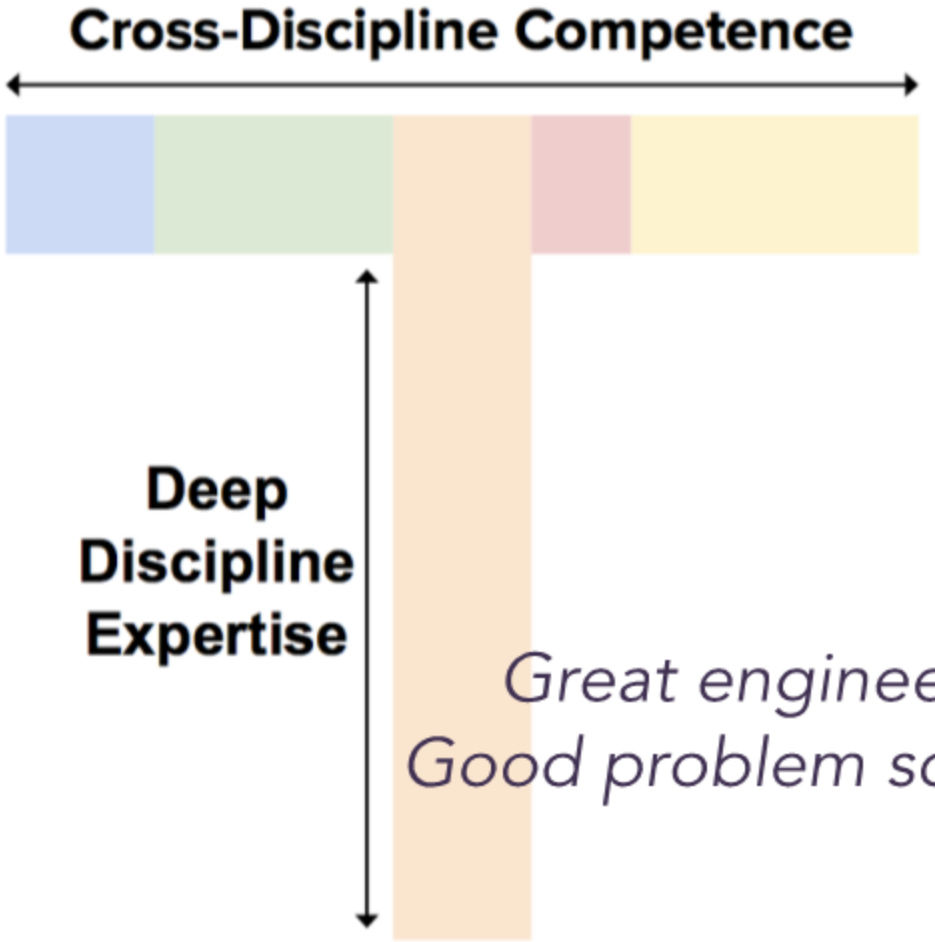
Bad engineers

Superficial
narrow knowledge

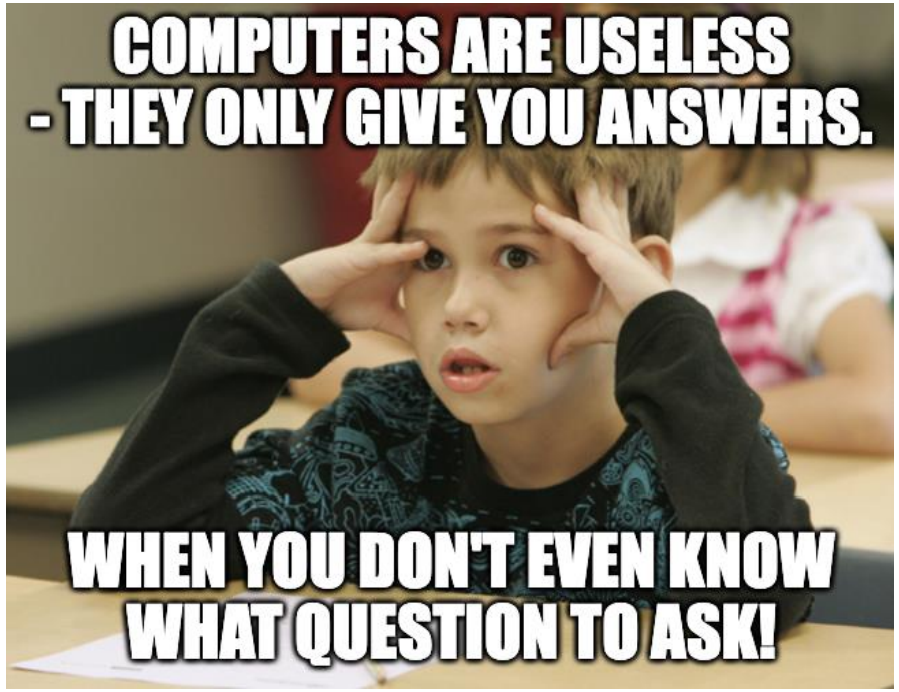


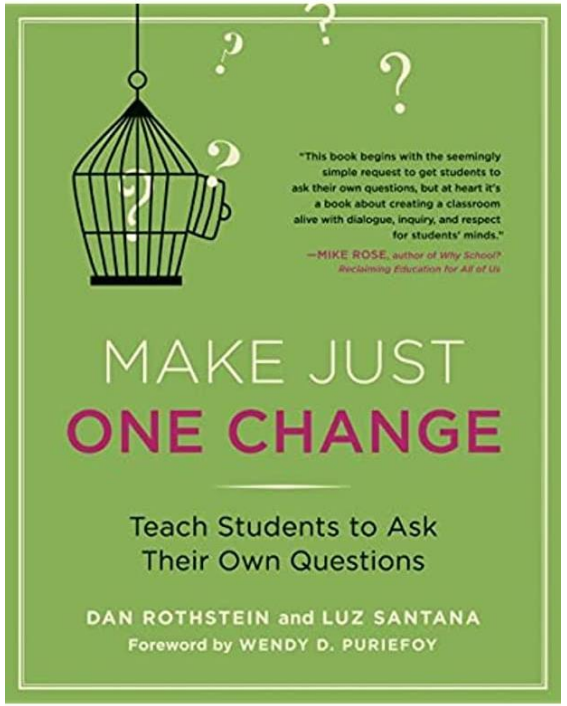
Good engineers
Bad problem solvers

Deep (but narrow) expertise



Enhancing IQ (inquiry quotient)

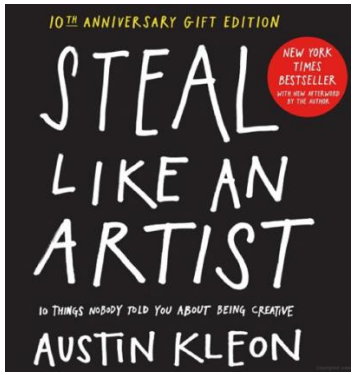


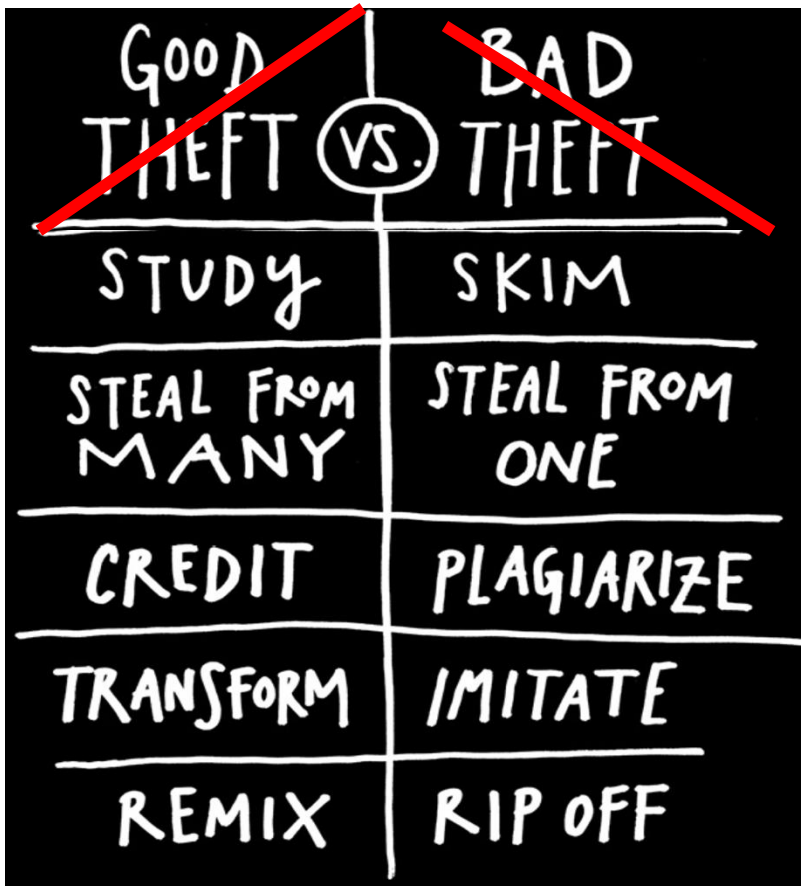


To engage students in meaning making and to develop self-directed learners, students should be encouraged to develop their own questions—
and to pursue the answers through active inquiry (i.e., to become researchers)

The importance of value creation

INSPIRATION PLAGIARISM





"Laws" of Human Intelligence Augmentation

[Aimed at Learners]

First Law 

When the AI is readily accessible, your work should be no worse than that produced by the AI alone.

(Embrace change and leverage AI)

Second Law 

(Corollary of Law 1)

When AI is there, and you don't want to be made irrelevant, maintain your competitive advantage

(Become **AI Proof**)

Third Law 

After becoming users of AI, don't let your inherent unaided capabilities drop

(**Don't outsource your thinking muscle allowing it to atrophy**; i.e., do no harm [*a la* Asimov; Hippocrates]

Towards Human Intelligence Augmentation

(What *Educators* Can Do)



Principle 1

The learning process should be centered around the student, rather than on the teacher, curriculum, or AI

(Be learner-centric and make your education human-centered)



Principle 2

AI should be utilized in a way that facilitates scalable iterative rapid ***formative feedback***

(Scale formative feedback using AI)

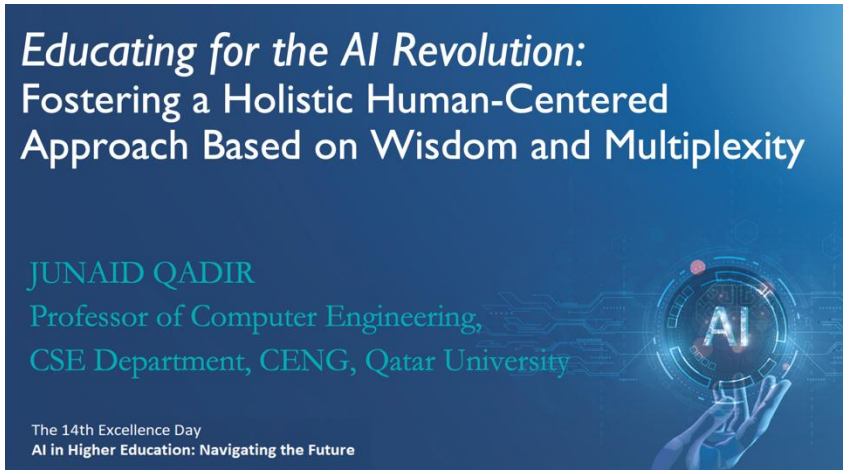


Principle 3

Teach the students the critical thinking skills necessary for judging and correcting the output of AI models

(Equip students with critical thinking to judge AI's output.]

Thank you!



We are actively engaged in R&D on technical problems related to developing trustworthy AI for higher education in Qatar.

For collaboration or suggestions, please reach out at:

jqadir@qu.edu.qa